

財政學 HW2

繳交期限：2024 年10 月9 日 23:59

1. Read the Summary of “The Rationale for Government Intervention” (the supplementary to Lecture I—Introduction 1). (不須繳交)

2. (Rosen 3.6) Imagine a simple economy with only two people: Augustus and Livia.

a. Let the social welfare function be

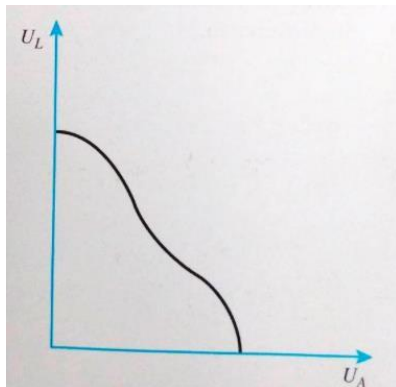
$$W = U_L + U_A$$

where  $U_L$  and  $U_A$  are the utilities of Livia and Augustus, respectively. Graph the social indifference curves. How would you describe the relative importance assigned to their respective well-being?

b. Repeat part *a* when

$$W = U_L + 2U_A$$

c. Assume that the utility possibilities curve is as follows:



Graphically show how the optimal solution differs between the welfare functions given in parts *a* and *b*.

3. (Rosen 3.9) Your airplane crashes in the Pacific Ocean. You land on a desert island with one other passenger. A box containing 100 little bags of peanuts also washes up on the island. The peanuts are the only thing to eat.

In this economy with two people, one commodity, and no production, represent the possible allocations in a diagram, and explain why every allocation is Pareto efficient. Is every allocation fair?

4. (Rosen 3.11) Suppose that Tang and Wilson must split affixed 400 pounds of food

between them. Tang's utility function is  $U_T = \sqrt{F_1}$  and Wilson's utility function is  $U_W = \frac{1}{2}\sqrt{F_2}$ , where  $F_1$  and  $F_2$  are pounds of food to Tang and Wilson, respectively.

- a. How much utility will Tang and Wilson receive if the food is distributed evenly between them?
- b. If the social welfare function is  $U_T + U_W$ , then what distribution of food between Tang and Wilson maximizes social welfare?
- c. If social welfare is maximized when they each obtain the same level of utility, then what is the distribution of food between Tang and Wilson that maximizes social welfare?

5. (Rosen 3.13) Suppose that Hannah's utility function is  $U_H = 3T + 4C$  and that Jose's utility function is  $U_J = 4T + 3C$ , where  $T$  is pounds of tea per year and  $C$  is pounds of coffee per year. Suppose there are fixed amounts of 28 pounds of coffee per year and 21 pounds of tea per year. Suppose also that the initial allocation is 15 pounds of coffee to Hannah (leaving 13 pounds of coffee to Jose) and 10 pounds of tea to Hannah (leaving 11 pounds of tea to Jose).

- a. What do the utility functions say about the marginal rates of substitution of coffee for tea?
- b. Draw the Edgeworth Box showing indifference curves and the initial allocation.
- c. Draw the contract curve on the Edgeworth Box. Explain why it looks different from the contract curves depicted in the text.
- d. Is the initial allocation of coffee and tea Pareto efficient?

6. (Rosen 3.14)

Indicate whether each of the following statements is true, false, or uncertain, and justify your answer.

- a. If everyone has the same marginal rate of substitution, then the allocation of resources is Pareto efficient.
- b. If the allocation of resources is Pareto efficient, then everyone has the same marginal rate of substitution.
- c. A policy change increases social welfare if, and only if, it represents a Pareto improvement.
- d. A reallocation from a point within the utility possibilities curve to a point on the utility possibilities curve results in a Pareto improvement.

7. (Hyman 2.1)

The following table shows how the total social benefit and total social cost of summer outdoor concerts in Central City vary with the number of performances.

| Number of Concerts | Total Social Benefit(\$) | Total Social Cost(\$) |
|--------------------|--------------------------|-----------------------|
| 1                  | 10,000                   | 5,000                 |
| 2                  | 15,000                   | 11,000                |
| 3                  | 18,000                   | 18,000                |
| 4                  | 20,000                   | 26,000                |
| 5                  | 21,000                   | 36,000                |

What is the efficient number of concerts?

8. (Hyman2.3)

A prominent senator has calculated the total social benefit of the current amount of space exploration at \$3 billion per year. The total social cost of space exploration is currently only \$2 billion. The senator argues that a net gain to society would result by increasing the amount of space exploration until total costs rise enough to equal total benefits. Is the senator's logic correct?